

1

Motivation

REWRITE THE FOUNDATION

- The paper argues that conventional ML is opaque internally and Transformer attention scales poorly with context.
- It draws inspiration from cortical semantic maps and electrophysiological signal propagation.
- Goal: component-level semantic interpretability and model size independent of sequence length.

2

SiFu

STATIC MAP + DYNAMIC FLOW

- Every vocabulary token maps to a dedicated graph node, analogous to a semantic cortical region.
- Bidirectional learned edges carry signals along token sequences; nodes and edges transform activation.
- Prediction becomes search for the candidate node receiving the greatest energy.

3

BriLLM

ENERGY-BASED GENERATION

- Each node is a GeLU neuron layer; each directed edge has a 32 x 32 matrix and bias.
- Positional encodings preserve order, while softmax weights combine signals from all previous nodes.
- Cross-entropy increases the energy of the correct next-token path relative to alternatives.

BRILLM / SiFu

Brain-Inspired
Large Language Model

Signal Fully-Connected Flowing | Work in Progress, 2025

Represent vocabulary items as explicit semantic nodes and generate by propagating signals through token-to-token edges, choosing the next node with maximum signal energy.

brain-inspiredsemantic graphsignal energy

4

Mathematics

SIGNAL INTEGRATION

- $E_{(L-1)} = \sum_k \text{softmax}(\alpha)_k e_k$
- $v^* = \operatorname{argmax}_v ||\text{GeLU}(W_{(u,v)} E_{(L-1)} + b_{(u,v)} + PE)||_2$
- Sparse training shares matrices for rare co-occurrences, reducing a fully connected graph by about 87-94%.

5

Evidence

EARLY FEASIBILITY ONLY

- Chinese/English Wikipedia runs use a 4K vocabulary, 32-token sequences and only 1.5K training steps.
- Sparse checkpoints contain 2.19B and 0.96B parameters; loss declines and training-text continuation is partly coherent.
- Held-out English/Chinese samples remain highly garbled, so GPT-1-level quality and AGI claims are not established.

6

Claims & limits

READ CRITICALLY

- Authors project 100-200B parameters for a 40K vocabulary and claim context-length-independent model size.
- The graph itself scales $O(\text{vocab}^2 \times \text{node_dim}^2)$; practical compute, long-context accuracy and multimodality are untested.
- This is an exploratory work-in-progress, not evidence of full interpretability, unbounded context or an AGI path.